

Metacognition and Confidence: A Review and Synthesis

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Keywords

metacognition, confidence, uncertainty, self-model, computation

Abstract

Determining the psychological, computational, and neural bases of confidence and uncertainty holds promise for understanding foundational aspects of human metacognition. While a neuroscience of confidence has focused on the mechanisms underpinning subpersonal phenomena such as representations of uncertainty in the visual or motor system, metacognition research has been concerned with personal-level beliefs and knowledge about self-performance. I provide a road map for bridging this divide by focusing on a particular class of confidence computation: propositional confidence in one's own (hypothetical) decisions or actions. Propositional confidence is informed by the observer's models of the world and their cognitive system, which may be more or less accurate—thus explaining why metacognitive judgments are inferential and sometimes diverge from task performance. Disparate findings on the neural basis of uncertainty and performance monitoring are integrated into a common framework, and a new understanding of the locus of action of metacognitive interventions is developed.

Contents

1. INTRODUCTION	242
2. SCOPE AND DEFINITIONS	243
3. PARADIGMS AND FINDINGS IN METACOGNITIVE NEUROSCIENCE	245
4. COMPONENTS OF A METACOGNITIVE JUDGMENT	247
4.1. Representing Uncertainty	247
4.2. Propositional Confidence	250
4.3. Global Broadcast and Communication	253
4.4. The Role of Self-Models	254
5. CONFIDENCE FORMATION AND THE PSYCHOLOGY OF METACOGNITION	256
6. REVISITING CURRENT CONTROVERSIES	257
6.1. Biases and Suboptimalities in Confidence	257
6.2. Sources of Domain-Generalizability	258
7. WHERE NEXT?	259
7.1. Searching for Common Computational Principles	259
7.2. From Local to Global Metacognition	260
7.3. Opportunities for Metacognitive Interventions	260
8. CONCLUSIONS	261

1. INTRODUCTION

Imagine you are revising for an upcoming exam in psychology. At various points leading up to the big day, you wonder whether you know the material well enough or not. Such an assessment might prompt further study, until those uncertainties are diminished and you feel more confident in being able to answer anything that is thrown at you. Before going into the exam hall, you nervously compare your chances of success with those of your friends. Later, after the exam is over, you think back over your answers, questioning whether the exam went well or could have gone better. These forms of self-evaluation are instances of metacognition—the capacity to reflect on, evaluate, and control mental function in a variety of useful ways.

These are examples of metacognition about memory, or metamemory for short; but metacognition operates over a range of domains. Consider a visit to the optician for a new pair of glasses. In a typical eye exam, you will be asked whether you are seeing the world more or less clearly through different lenses. This is a metacognitive judgment about your perceptions: The world is not blurry, but a limit on your visual acuity makes it seem so.

It is hopefully clear from these two examples that the accuracy of metacognition—whether or not our self-evaluative judgments match up with the reality of cognitive or physical performance—is central to adaptive behavior. If I think that my knowledge about a topic is secure when it is in fact shaky, I might put down the books and go out with my friends, only to be in with a nasty shock on exam day. Similarly, if we are unable to realize when our vision (or hearing, or memory) is failing, we will be unable to take steps to correct for physical or cognitive limitations. As such, metacognitive dysfunction has been highlighted as a key source of maladaptive behavior in educational, clinical, and societal contexts (Flavell 1979, Hoven et al. 2019, Rollwage et al. 2018).

Effectively estimating our uncertainty or confidence in a range of cognitive processes, and whether or not such confidence judgments track objective performance (known as metacognitive

sensitivity), is therefore central to effective metacognition (Nelson & Narens 1990). Miscalibrated confidence in success can lead to failure, even when our natural aptitude is more than adequate. Recently, there has been a surge of interest in the neuroscience of uncertainty and confidence, leading to a marriage of computational work in cognitive science with human neuroimaging studies and animal models of metacognitive judgments (Meyniel et al. 2015, Pouget et al. 2016). Partly because these fields were steeped in the methods of psychophysics, and partly because of the cross-species tractability of perceptual paradigms, the late 2000s saw the emergence of the field of perceptual (largely visual) metacognition, with a strong focus on the neural and computational underpinnings of confidence judgments (Rahnev 2021).

However, the rapid rise of this research program brings with it a set of pressing conceptual challenges. The neuroscience of confidence has tended to focus on the mechanisms underpinning subpersonal phenomena such as the representation of uncertainty in the visual or motor system, often in tightly controlled laboratory tasks. Conversely, metacognition researchers are interested in personal-level beliefs and knowledge in real-world settings: Why do I think that I performed poorly on the exam? How do I recognize when I might have made a poor decision? Why is a patient with Alzheimer's disease unaware of their memory failures? How do children form beliefs about what they know and do not know?

In this article I aim to provide a road map for bridging this divide. Metacognition and confidence researchers are natural allies but have often been uneasy bedfellows, with the latter thinking that the former are overcomplicating things, and the former thinking that the latter are riding roughshod over the richness of metacognition by reducing it down to its computational primitives. I suggest that one solution to understanding the role of confidence in real-world metacognition is to focus on a particular class of confidence computation: propositional confidence. Propositional confidence is confidence in one's own (hypothetical) decisions or actions, which include covert propositions (e.g., "I think I will remember this word"; see **Figure 1**). The most important idea, building on the work of Pouget et al. (2016), is that propositional confidence can be distinguished from a myriad of other confidences or uncertainties that are inherent to perception, cognition, and action, although the latter often inform the former (Meyniel et al. 2015). Propositional confidence is also affected by the observer's models of the world and their cognitive system, which may be more or less accurate, thus explaining why metacognitive judgments are inferential and sometimes diverge from task performance.

2. SCOPE AND DEFINITIONS

The terms metacognition and confidence can take on different meanings in different research fields, and so it is useful to spend some time providing explicit definitions.

By metacognition, I refer to the class of mechanisms that allow us to form beliefs about other mental operations. Such beliefs (the monitoring aspect of metacognition) can then be harnessed for self-regulation (metacognitive control) and/or for communicating metacognitive assessments to others. Metacognition is a part of the wider set of human executive functions, although it is conceptually and empirically distinct from fluid intelligence: It is possible (and indeed common) to evaluate the operation of classical executive functions, for instance, reflecting on whether a solution to a logical puzzle was in fact appropriate (Ackerman & Thompson 2017). The accuracy of such reflective judgments shares variance with other forms of metacognitive sensitivity rather than variance in IQ (Mazancieux et al. 2020, Rouault et al. 2018a). Finally, metacognition also intersects with the literature on cognitive control, although again with only partial overlap. Cognitive control typically refers to the set of functions that encode and maintain a representation of the current (first-order) task. For instance, in Miller & Cohen's (2001) classic model of cognitive control, prefrontal cortex provides contextual signals to bias or route sensory information to

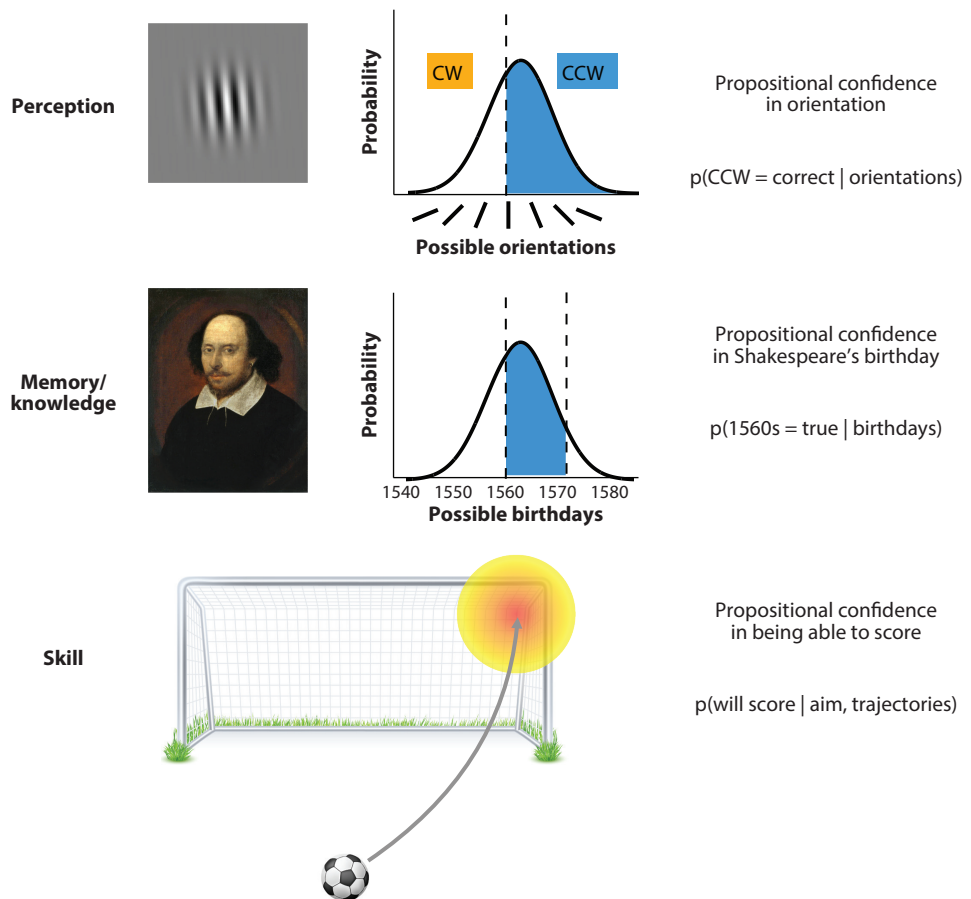


Figure 1

Metacognitive judgments can be formalized as estimates of propositional confidence across a range of domains and timescales. Abbreviations: CCW, counterclockwise; CW, clockwise.

establish the right mapping between inputs, internal states, and outputs. All of this machinery can be considered as being part of the same (context-sensitive) first-order system. We can then apply metacognitive mechanisms to monitor task performance and subsequently increase our reliance on cognitive control (Norman & Shallice 1986). The literature on error correction and performance monitoring has often been lumped together with the literature on cognitive control, but here it would also fall under the rubric of metacognition research.

By confidence, I mean the degree of belief one has about the likely success of a variety of mental operations. Thus, confidence refers here to propositional confidence—a feeling of surety about one's abilities, judgments, or ideas. Confidence also has a more general meaning as a synonym for probability—e.g., when ascribing a high probability (high confidence) to the fact that the sun will rise tomorrow. Such probabilities apply to external quantities, independently of an observer. To add to the confusion, it is also possible that the brain itself uses probabilistic computation in a range of processes, including the formation of feelings of confidence! To try to avoid confusion here, I will follow Pouget et al. (2016) and reserve the term confidence to refer to propositional

confidence in a (mental or physical) action, and I will use the term certainty (or its converse, uncertainty) to refer to degree of belief in other quantities.

I aim to bridge the work on subpersonal representations of uncertainty, personal-level feelings of confidence, and the operation of metacognition more broadly. This necessarily means being selective in the empirical literature that is most helpful in illuminating those relationships. As such, there are a number of topics that fall outside the scope of this review, given limited space. These are the development of metacognition; comparative research on animal metacognition; links between metacognition, mental health, and ageing (but see the sidebar titled Individual and Group Differences in Section 6.2); and interpersonal and intrapersonal functions of metacognition.

The outline of the article is as follows. In Section 3 I provide a brief overview of core findings in metacognitive neuroscience that motivate the current synthesis. Section 4 deconstructs the different components of a personal-level metacognitive judgment and reviews the evidence for distinct components, with a particular focus on neuroscience. An important concept here will be the notion of a reference frame. We can talk of uncertainty about things in the world, such as sensory uncertainty about the orientation of a line or the frequency of a sound. This is uncertainty in a world-centered reference frame. As we have seen, however, we can also talk of confidence in our own propositions or actions; this is now uncertainty in a self-centered reference frame. In Section 4 I turn to how such signals are read out or broadcast in a format that is useful for guiding behavior and communication to others, before evaluating in Section 5 the role that model-based computation plays in providing contextual knowledge for metacognition.

In Section 6 I discuss how current controversies in metacognition research can be reevaluated in light of this framework—in particular, the origin of biases and suboptimalities in metacognition, how to arbitrate between computational models of confidence, and whether or not we should consider metacognition as a domain-general resource. I close in Section 7 by highlighting some future directions that are motivated by this framework—in particular, searching for common computational principles across different task domains, extending models of local confidence to understand the formation of metacognitive knowledge over longer timescales, and identifying the best routes for interventions on metacognition.

3. PARADIGMS AND FINDINGS IN METACOGNITIVE NEUROSCIENCE

A range of behavioral paradigms investigating different types of metacognitive judgment have been devised, often originating in work on metamemory and ranging from prospective judgments of learning to retrospective confidence estimates in recall (Metcalf & Shimamura 1994). All paradigms, however, have in common that subjects are being asked to evaluate their (future or past) performance on another task. As we will see, such evaluations are naturally cast as judgments of propositional confidence in the success of other mental operations. In humans, these judgments are usually explicit and instructed: Subjects are provided with a button or scale on which to indicate their confidence or are asked, in the confidence forced-choice paradigm, to pick from a pair of decisions the one they feel most confident about (Mamassian & de Gardelle 2022). In animal metacognition research, confidence estimates are elicited using a variety of learned second-order contingencies such as opting out of a decision, waiting for a reward that is contingent on first-order task performance, and so on (Kepecs & Mainen 2012). These so-called implicit measures of metacognition have recently found their way into innovative studies of infant metacognition, where explicit confidence elicitation is less straightforward (Goupil & Kouider 2016).

When we have data on a series of metacognitive judgments over time, we can examine the statistical association between behavioral performance and metacognition. Intuitively, if you are confident when you are right, and less confident when you are wrong, then you can be ascribed

MEASUREMENT OF METACOGNITION

Measures of metacognition in experimental tasks seek to estimate the statistical relationship between confidence judgments and objective performance, known as metacognitive sensitivity. A central challenge in this endeavor is to ensure that metrics of metacognitive sensitivity are unconfounded by other influences. For instance, simple correlations between accuracy and confidence not only depend on metacognitive sensitivity but also are affected by performance and metacognitive bias (Fleming & Lau 2014). The meta- d' model offers a performance-controlled metric of metacognitive sensitivity by estimating the level of first-order performance (d') that would have given rise to the observed confidence data under a signal detection theoretic model (Maniscalco & Lau 2012). The ratio meta- d'/d' thus provides a performance-controlled metric of metacognitive capacity (often referred to as metacognitive efficiency). However, the assumption that meta- d'/d' is fully independent of metacognitive bias and performance has been challenged (Guggenmos 2021, Xue et al. 2021). Alternative model-free approaches assess the mutual information between performance accuracy and confidence reports (Dayan 2022) or quantify the change in psychometric function slope as a function of confidence (de Gardelle & Mamassian 2014, De Martino et al. 2013).

a high degree of metacognitive sensitivity (Fleming & Lau 2014). Another relevant summary statistic for investigations of metacognition is metacognitive bias (also known as calibration or overconfidence), that is, the extent to which subjects tend to report higher or lower confidence relative to long-run performance. One challenge is to ensure that measures of metacognitive sensitivity are unconfounded by other factors, including task performance, metacognitive biases, and response times (see the sidebar titled Measurement of Metacognition).

With these metrics in place, two lines of work have emerged in metacognitive psychology and neuroscience over the past few decades. The first has sought to catalog both individual differences and interventions—either experimentally controlled or naturally occurring in the form of brain damage or disorder—that affect metacognition without affecting first-order task performance. A second line of work has focused on the psychological, computational, and neural basis of confidence formation across a number of different task domains in both humans and animal models. These research programs on individual differences and confidence formation naturally reinforce one another, as new discoveries about the formation of confidence can shed light on the origins of individual and group differences, and identifying individual and group differences in metacognitive efficiency provides hints about where to look for sources of noise or suboptimality in confidence formation.

Classical work in the cognitive psychology of metamemory has identified a range of cues that may affect confidence judgments but are unrelated to first-order performance. For instance, when attempting to recall a difficult-to-retrieve item, the extent to which we can recall information related to the target (cue accessibility) predicts how confident we are of being able to recognize the target (Koriat 1993). A number of these influences on metamemory judgments have been studied in depth—including target accessibility, fluency at encoding and retrieval, and response time—leading to the broad proposal (which we return to below) that metacognitive judgments are inferential in nature and draw on a range of helpful and unhelpful cues to performance (Metcalfe & Shimamura 1994, Nelson & Narens 1990). Within the field of metaperception research, studies have documented dissociations between confidence and accuracy as a function of attention (e.g., Wilimzig et al. 2008), variability in perceptual evidence (e.g., Spence et al. 2016, Zylberberg et al. 2014), asymmetries in the processing of supporting and disconfirming evidence (e.g., Miyoshi & Lau 2020, Zylberberg et al. 2012), and response times (e.g., Kiani et al. 2014).

Pioneering neuropsychological investigations of patients with frontal lobe damage have identified a key role for human prefrontal cortex in supporting metacognitive capacity, often on memory tasks (see Fleming & Dolan 2012, Pannu & Kaszniak 2005 for reviews). The importance of prefrontal cortical function in metacognition has been supported by recent studies in both humans and animals. Changes in confidence formation and metacognition (but not first-order task performance) are observed following temporary disruption or lesions to rostrolateral prefrontal cortex (Brodmann areas 46 and 10) in humans and monkeys (Fleming et al. 2014; Kwok et al. 2019; Miyamoto et al. 2017, 2018; Shekhar & Rahnev 2018), and confidence-related behavior is impaired following inactivation of orbitofrontal cortex (OFC) in rodents (Lak et al. 2014). Individual differences in perceptual metacognitive sensitivity have been similarly linked to variation in the structure and function of human anterior prefrontal cortex (Allen et al. 2017, Baird et al. 2013, Fleming et al. 2010, McCurdy et al. 2013). This picture of a unitary prefrontal correlate of metacognition has been nuanced with observations in humans that distinct brain systems may predict metacognitive sensitivity in perception and memory tasks (Baird et al. 2013, Fleming et al. 2014, McCurdy et al. 2013, Ye et al. 2018), and that connectivity between prefrontal cortex and other brain areas is important for metacognitive capacity (Baird et al. 2013, 2015; De Martino et al. 2013; Zheng et al. 2021).

Finally, a number of studies in both human and animal models have sought to relate variation in subjective confidence reports, or confidence-related behaviors, to changes in neural activity measured either with single-unit recordings or with mass univariate analyses of neuroimaging data. Many of these studies are discussed in more detail in subsequent sections. For now, it is sufficient to say that the field has cataloged a wide variety of confidence-related neural signals (Walker et al. 2023), with the functional anatomy of metacognition becoming both richer and more complex. Imposing order on these findings is one of the goals of this review: How can we square the often striking dissociations between performance and metacognition observed in lesion studies with the multiplicity of neural representations of uncertainty and confidence? In the remainder of this article I develop the computational components of a metacognitive judgment, beginning with a theoretical perspective and then turning to consider the behavioral and neuroscience evidence for each component.

4. COMPONENTS OF A METACOGNITIVE JUDGMENT

4.1. Representing Uncertainty

Metacognitive assessments refer to one's degree of certainty or uncertainty about a particular mental operation. It is thus natural that the uncertainty inherent to neural representations of sensory features should be highly relevant to metacognition. When a doctor views an X-ray, the incoming visual information may be consistent with a number of different interpretations (both of simple features such as lines and edges and of more global properties such as the presence or absence of a tumor). It is increasingly recognized that uncertainty is inherent to all stages of neural computation and that optimal behavior requires sensitivity to such uncertainty. For instance, when combining information from two different sensory modalities, the normative (Bayesian) solution is to weight the two sources inversely according to their respective uncertainties.

Within perceptual systems, different competing theoretical schemas have been proposed for how the brain represents uncertainty about particular quantities. Consider a judgment of the orientation of a low-contrast grating (see **Figure 2**). The sensory data underdetermine the true orientation, leading to uncertainty in the internal representation of orientation z (note that this uncertainty is subjective uncertainty in the representation rather than noise in the stimulus, although the latter may affect the former). We can denote such uncertainty as a (posterior) probability

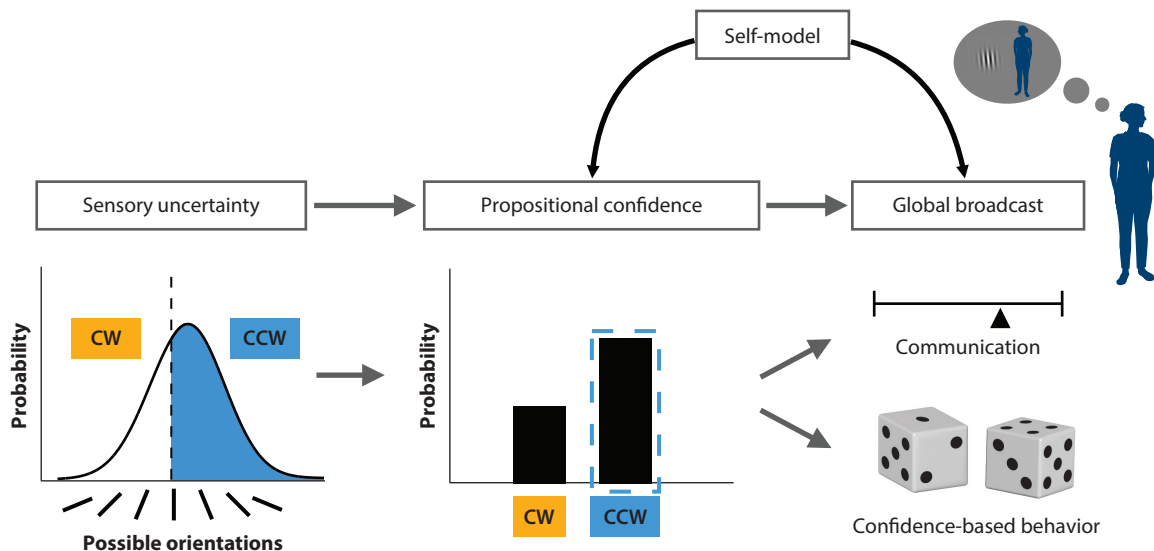


Figure 2

Graphical illustration of the components of a perceptual metacognitive judgment. A generative model defines how an observer forms a belief about the state of the world—here, the orientation of the stimulus—from a noisy sensory measurement. This belief over possible orientations is associated with sensory uncertainty and is converted into propositional confidence conditional on a categorical decision—here, whether the stimulus is tilted clockwise (CW) or counterclockwise (CCW). A propositional confidence estimate is globally broadcast for communication or usage in confidence-based behaviors (for instance, for guiding risk-sensitive decision making). Background beliefs about a range of factors influencing self-performance are furnished by a self-model and influence the formation of metacognitive judgments.

distribution around the most probable orientation. Under a probabilistic population coding model, neurons encode parameters of probability distributions, with different neurons tuned to different stimulus features (e.g., its orientation or color), such that a population of such neurons represents a probability distribution over features, given a sensory measurement (Ma et al. 2006). Alternative schemes include sampling-based accounts, in which samples from a distribution are accumulated over time in the form of spikes, and summary-statistic accounts, in which neuromodulators or other aspects of brain activity carry uncertainty-related information (Fiser et al. 2010, Yu & Dayan 2005).

For our current purposes, it is sufficient to note that a number of theoretical accounts propose that neural representations come along with an implicit representation of the certainty with which that representation is held. Such distributional uncertainty is thought to be encoded at a number of different levels, from perception to cognition and action. As concrete examples, a population of neurons in V1 might (implicitly) carry information about the uncertainty of the orientation of a low-contrast bar, a population of neurons in auditory cortex may carry information about the uncertainty of the frequency of a tone in noise, and so on. These examples hopefully make clear that the brain can, and likely does, track uncertainty in a whole host of quantities. Bayesian theories of brain function additionally propose that such uncertainties determine the appropriate weighting of messages passed up and down a cognitive hierarchy. Following Meyniel et al. (2015), I refer to these uncertainty signals as implicit or distributional uncertainty, but such estimates may also be transformed into scalar summary signals (e.g., a scalar signal of sensory uncertainty signaled by the level of a particular neuromodulator).

A wide range of studies indicate that subjects take into account uncertainty in their behavior, including in experiments on perception, learning, memory, and motor control (Kersten et al.

2004, Meyniel et al. 2015, Trommershauser et al. 2008). Some of the most robust evidence for the representation and use of uncertainty comes from the literature on cue combination in multisensory integration. If subjects are asked to combine information across two sensory modalities, the weights they put on the two sources of information are inversely proportional to their uncertainty and approach the predictions of an ideal Bayesian observer (e.g., Ernst & Banks 2002). Similarly, in the motor domain, subjects are sensitive to uncertainty in movement production (e.g., the dispersion of rapid pointing movements) and use this information to alter their movement strategies to avoid risky actions (Trommershauser et al. 2008).

Such studies, however, do not tell us whether uncertainty is used to inform metacognition. A number of studies have presented evidence that confidence judgments are sensitive to the variability in perceptual evidence, although sometimes to a greater or lesser degree than predicted by an ideal observer model (Boldt et al. 2017, Spence et al. 2016, Zylberberg et al. 2014). Other work has revealed how people adjust their confidence criteria in the face of changing stimulus uncertainty (Adler & Ma 2018, Aitchison et al. 2015, Denison et al. 2018). However, such results rely on comparing model fit across multiple trials and admit heuristic accounts of how uncertainty affects confidence. Establishing that uncertainty on individual trials is used to inform confidence judgments has proven more difficult.

Neuroscience evidence makes a stronger case for the idea that uncertainty estimates inform confidence judgments. Kiani & Shadlen (2009) found that activity in lateral intraparietal cortex (area LIP) in the monkey brain accumulated evidence for particular choice options and, when such activity was of intermediate strength, led the monkeys to opt out of their choice (a nonverbal marker of low certainty about either motion direction). Importantly, variability in LIP firing rates predicted the opt-out choice even when stimuli were held fixed, drawing a link between neural and behavioral markers of certainty about motion direction. Note that such activity is in a world-centered reference frame (reflecting certainty about the mapping between the stimulus and potential responses) rather than in a self-centered reference frame. However, such a representation naturally supports prospective propositional confidence estimates (e.g., “How confident am I in choosing A or B, conditional on the evidence that I have gathered so far?”). The opt-out task is thus an ambiguous case: It can be solved by relying on world-centered uncertainty estimates or self-centered (metacognitive) confidence estimates, and it is hard to tell which ones are at play based on behavior or neural data alone.

Geurts et al. (2022) asked human participants to estimate the orientation of a tilted grating and judge their confidence that their estimate was accurate. Within a Bayesian framework, it was expected that the more precise the representation of orientation in visual cortex (the smaller the posterior uncertainty), the larger the propositional confidence in the tilt estimate. This was the case in the subjects’ behavior. The authors then used a machine learning approach to decode trial-by-trial uncertainty in the representation of particular orientations from fMRI voxel patterns within visual cortex. They found that reported confidence was negatively correlated with such a readout of uncertainty, even when the stimulus was held fixed.

There is thus good evidence that (a) the brain tracks uncertainty about a wide range of quantities and (b) such uncertainty informs metacognitive judgments. It remains unclear how and whether a similar scheme is maintained beyond sensory representations—for instance, when judging confidence in being able to remember something. Recent fMRI evidence suggests similar population-level representations of uncertainty in visual working memory (Li et al. 2021), and single unit activity in the human hippocampus predicts retrieval confidence levels (Rutishauser et al. 2015). Sampling schemes offer another potential solution, allowing probability distributions over internal states to be formed by drawing samples from internal models (Fiser et al. 2010).

4.2. Propositional Confidence

Representing certainty or uncertainty in a self-centered frame of reference—what I refer to as propositional confidence—is the foundation of metacognitive judgments. Computationally, this can be achieved by transforming an internal (sensory or mnemonic) representation z into an estimate of confidence in taking an action based on z (see the sidebar titled Computing Propositional Confidence). For a Bayesian observer, if z indicates a probability distribution (posterior) over possible orientations (see **Figure 2**), and the observer's task is to say whether the orientation is clockwise or counterclockwise (a binary variable, d), a confidence judgment can be derived from computing $p(d = a|z, a)$ —that is, the probability that action a picked out the correct world state d , given z . In a situation where one's action is based solely on z , propositional confidence is a nonlinear transformation of z . However, if there are additional sources of decisional or metacognitive noise, or if additional information arrives after committing to a decision, then propositional confidence should also be affected by these factors (Fleming & Daw 2017). In all these cases, propositional confidence should be closely informed by estimates of uncertainty reviewed in the previous section. The upshot is a confidence estimate in the frame of reference of the accuracy of one's own judgments—a self-related frame of reference.

COMPUTING PROPOSITIONAL CONFIDENCE

Consider a visual perceptual task in which the decision maker should classify the orientation of a stimulus s as clockwise (CW) or counterclockwise (CCW) relative to some arbitrary boundary m (Bang & Fleming 2018). On a single trial, the observer makes a sensory measurement X_i . The posterior over possible orientations s is then

$$p(s|X_i) \propto p(X_i|s)p(s).$$

Because measurements are affected by noise, for a single stimulus, the measurement X_i is a bit more or a bit less than the true s . This can often be controlled by the experimenter, for instance, by adjusting the contrast of a grating or the coherence of a patch of randomly moving dots. Under greater noise, the likelihood of s becomes wider (the first term on the right-hand side), because the measurement is potentially consistent with a wider range of true orientations. Assuming the prior stays constant, this also leads to a more uncertain posterior over s (the left-hand side).

The observer now has an internal belief with some sensory (or mnemonic) uncertainty attached to it, but they still need to act on this information—in this example, by saying whether the orientation is CW or CCW to the boundary. Doing so requires specifying which actions are possible (mappings from s to a) and the cost or reward associated with each. Here it is useful to specify an intermediate variable that captures relevant parts of the stimulus space: d is CW when $s < m$ and CCW when $s > m$. In the case of a simple perceptual decision-making task that rewards correct decisions, the cost function $C(d, a)$ is 1 when $a = d$ and 0 otherwise. We can now define a new form of certainty about possible actions (see **Figure 2**) as

$$p(d_{CW}|\hat{s}) = \int_{-\infty}^m p(s|X_i),$$

$$p(d_{CCW}|\hat{s}) = \int_m^{\infty} p(s|X_i),$$

where $\hat{s}$ indicates the observer's estimate of s .

Once we have committed to a potential action (an action that will occur or has occurred), we can use the above probabilities to compute the probability that the action was correct [i.e., that $C(d, a) = 1$]:

$$p(a = d|\hat{s}).$$

This quantity is what I refer to as propositional confidence (Pouget et al. 2016).

It is natural to think of such a change in reference frame as being retrospective: I process some information, make a decision, and then reflect on whether my decision was correct. Indeed, as we will see, postdecisional processing is an important empirical signature of this change in reference frame. However, propositional confidence can also be prospective. Based on some uncertain information, I might estimate the likelihood that a hypothetical decision based on that information would be correct. Such prospective judgments can apply to propositions rather than individual actions—for instance, the proposition that “I will remember this particular word” or “I will score a goal” (see **Figure 1**). These prospective confidence estimates may therefore underpin classical judgments of learning or aspects of self-confidence about ability.

More recently, defining decision confidence as a Bayesian probability of being correct has been challenged on both empirical and theoretical grounds. Empirically, confidence closely tracks the probability of making a particular choice, rather than objective notions of accuracy. For instance, if choice probability is biased by perceptual illusions, confidence often follows suit (Caziot & Mamassian 2021, Gallagher et al. 2019). Theoretically, it is also hard to define notions of accuracy for subjective decisions, such as value-based choices or aesthetic preferences—and yet we can still evaluate confidence in such decisions (De Martino et al. 2013, Lebreton et al. 2015). Instead, a more general computational definition posits that propositional confidence reflects the probability of making a self-consistent choice across multiple presentations of the same decision problem (Boundy-Singer et al. 2023, Caziot & Mamassian 2021, Koriati 2012).

There have been two broad approaches to studying the behavioral and neural basis of propositional confidence. One is to simply ask for subjective reports of confidence about a future or past decision. These confidence judgments are higher for objectively correct decisions than for incorrect ones, showing sensitivity to performance, albeit often corrupted by additional metacognitive noise (Shekhar & Rahnev 2021). Convergent findings have emphasized the importance of human prefrontal cortex for the fidelity of propositional confidence estimates, with a meta-analysis revealing that activity in medial and lateral prefrontal cortex, precuneus, and ventral striatum covaries with judgments of confidence in memory and perceptual tasks (Vaccaro & Fleming 2018).

A second approach harnesses statistical signatures of confidence in a self-centered (decisional) frame of reference. A prominent signature here is the folded X pattern: When confidence is plotted against objective measures of signal strength (the inverse of decision difficulty), propositional confidence should increase with signal strength for correct trials and decrease with signal strength for error trials. The idea here is that, while errors on easier trials will be less frequent, those that do occur will be accompanied by significant evidence against the chosen option, leading to lower confidence. This pattern is seen in both human and animal confidence data (Sanders et al. 2016) and has been used as a marker of confidence-related physiological and neural signals (Urai et al. 2017). In a seminal study, Kepecs and colleagues found that neurons in rodent OFC showed statistical signatures of confidence in an odor discrimination task (Kepecs et al. 2008). Confidence signatures in OFC predict confidence-related behavior (waiting for a reward, conditional on performance) and generalize across both auditory and olfactory decisions (Masset et al. 2020), with inactivation of this brain area impairing metacognition but not performance (Lak et al. 2014). Rodent OFC is therefore a candidate neural substrate for propositional confidence.

A similar approach was adopted by Bang & Fleming (2018) in humans, in an fMRI study which manipulated both a proxy for sensory uncertainty (motion coherence) and the difficulty of the choice. Human participants viewed a random dot motion stimulus that indicated a particular direction around the circle with a given uncertainty, controlled by coherence. They then saw a decision boundary appear before participants were asked to decide whether the motion direction was clockwise or counterclockwise of the boundary. This design dissociates propositional confidence in a choice (which is affected by both sensory uncertainty and decision difficulty) from

sensory uncertainty (though here uncertainty was not directly assayed from neural representations and was confounded with stimulus properties). Whereas sensory uncertainty (motion coherence) was related to activity in extrastriate visual and parietal cortex (notably, areas MT+ and bilateral intraparietal sulcus, a human homologue of LIP), signatures of propositional confidence were instead observed within perigenual anterior cingulate cortex (pgACC), part of the ventromedial prefrontal cortex (vmPFC).

A complementary perspective on the neural basis of propositional confidence is provided by the literature on error monitoring, which has typically used speeded response-conflict tasks to induce response errors under time pressure. A canonical finding is that posterior medial frontal cortex (pmPFC) neurons covary with error commission in the absence of feedback, generating an error-related negativity (ERN) at the scalp surface (Desender et al. 2021). The ERN peaks approximately 100 ms after the erroneous action and arises before any feedback is given about the accuracy of the response. In animal models, postdecisional firing rates of neurons in prefrontal cortex and dopaminergic midbrain have also been shown to covary with choice correctness before explicit feedback is given (Kepecs et al. 2008, Middlebrooks & Sommer 2012, Tsujimoto et al. 2010). Within a reinforcement learning framework, one perspective on such signals is that they reflect proxies for reward prediction errors driven not by external feedback but by internal levels of choice confidence (Guggenmos et al. 2016, Lak et al. 2017).

More recently, it has been argued that postdecisional accumulation of evidence facilitates the formation of propositional confidence (Desender et al. 2021). The idea here builds on classical evidence accumulation frameworks positing that samples of sensory information are accumulated over a few hundred milliseconds before hitting the bound for one or other choice option. Such models have been highly successful in accounting for choice and response time behavior in a variety of decision scenarios, and neural correlates of evidence accumulation signals have been identified in humans and animals. Moreover, as we saw above, the dynamics of evidence accumulation within the choice period provide a neural representation of uncertainty that can be used to inform confidence (Kiani & Shadlen 2009). Pleskac & Busemeyer (2010) additionally proposed that this evidence accumulation process may continue after a decision has been made, informing estimates of decision confidence and potentially leading to changes of mind (Resulaj et al. 2009, van den Berg et al. 2016a).

Postdecisional processes may either continue to accumulate sensory evidence for and against available choice alternatives (world-centered reference frame) or accumulate evidence about the accuracy of the preceding choice (self-centered reference frame). Murphy et al. (2015) found that the ramping-like characteristics of a centroparietal electroencephalogram (EEG) signal, the Pe, was consistent with postdecisional evidence accumulation in a self-related reference frame. The postdecisional build-up rate of this signal was proportional to the speed of subjective error detection, and it reached a constant amplitude at the point of detection that was independent of error-detection response time. Interestingly, the Pe signature is similar to the centroparietal positivity (CPP) that has been linked to predecisional evidence accumulation in a world-centered reference frame. This suggests that the CPP and the Pe may reflect a general evidence accumulation circuit that can flexibly adapt reference frames in the service of both first-order performance and metacognition. Boldt & Yeung (2015) found that the Pe amplitude also predicts graded ratings of confidence in choice, highlighting how this accumulation signal goes beyond all-or-nothing error detection.

These studies investigated endogenous postdecisional accumulation of evidence. It is also possible to experimentally manipulate the availability of postdecisional information. Computationally, injecting additional postdecision evidence should promote the folded X pattern in confidence ratings, due to a greater opportunity for gaining evidence against an incorrect decision. In a study

of random dot motion discrimination, providing stronger postdecision evidence indeed led to a stronger folded X pattern in confidence ratings (Fleming et al. 2018). This folded X signature was observed in the fMRI activity of pMFC, consistent with this region (negatively) accumulating evidence in a frame of reference of choice accuracy and providing a computational bridge between studies on confidence and on error monitoring.

4.3. Global Broadcast and Communication

For propositional confidence to be useful to guide flexible behavior, it should be broadcast to a number of different consumer systems (Baars 1993). This would allow different propositional confidences to be compared in a common frame of reference—allowing the agent to decide, for instance, that they are more likely to be successful in judgments of one or other task or sensory modality (Aguilar-Lleyda & de Gardelle 2021). The global broadcast of confidence can also be used as a learning signal in lieu of external feedback—allowing agents the online detection of errors and consequent adjustments to behavior (Guggenmos et al. 2016). Interestingly, propositional confidence may emerge in parallel to the decision (or proposition) itself and may be used to shape the ongoing decision process—for instance, controlling the termination of evidence accumulation (Balsdon et al. 2020) or guiding the next step in a sequential decision (van den Berg et al. 2016b). Finally, global broadcast of propositional confidence is important for the public sharing of metacognitive representations in group settings: We might say to a colleague, “I believe this is the right thing to do,” thereby influencing the course of the group’s decision (Bahrami et al. 2010, Shea et al. 2014). Mappings between private feelings of confidence and public utterances lead to additional computational considerations. In a collaborative context, it is important to align the distribution of our confidence statements with those of others to avoid dominating a group interaction (or being dominated ourselves; Bang et al. 2017). However, if we wish to strategically influence the group, it might be advantageous to overstate (or understate) our public confidence (Hertz et al. 2017).

Global broadcast is proposed to covary with conscious awareness of a range of mental content, including metacognitive representations (Dehaene et al. 2017). This implies that forms of propositional confidence that remain restricted to a particular sensorimotor pathway and are not globally shared may underpin nonconscious forms of metacognition (Charles et al. 2013, Logan & Crump 2010). We may also consciously experience other forms of perceptual uncertainty beyond propositional confidence (Morrison 2016), and such uncertainty estimates may themselves affect what content is globally broadcast (Shea & Frith 2019).

Behaviorally, elegant work has shown that people are able to estimate and compare propositional confidence about decisions made in two different sensory modalities, indicating that domain-specific confidence estimates can be broadcast and shared (de Gardelle et al. 2016). There is also emerging evidence that metacognitive capacity (measured as the noise in metacognitive judgments relative to performance) covaries across perceptual and cognitive tasks, suggesting the existence of a global resource that is leveraged to monitor self-performance (Boundy-Singer et al. 2023, Mazancieux et al. 2020, Rouault et al. 2018b).

A common currency for confidence may be supported by modality-independent confidence signals in rodent (Masset et al. 2020) and human (Morales et al. 2018) prefrontal cortex. Recently, an impressive study conducted single-unit recordings in human neurosurgical patients performing two distinct tasks in which errors were relatively common (Fu et al. 2022). At the population level, pMFC cells formed a high-dimensional representation that allowed simple linear decoders to read out both domain-general error signals and, simultaneously, to differentiate domain-specific aspects of performance monitoring, such as the task and type of response conflict that gave rise to the error.

Performance monitoring signals are sensitive not only to the objective act of making an error but also to subjective error awareness (Nieuwenhuis et al. 2001) and decision confidence (Boldt & Yeung 2015), albeit with some intriguing dissociations that may indicate specific roles in global broadcast. The Pe (described in the previous section as being a candidate for postdecisional evidence accumulation) has been linked to error awareness and shown to covary with subjective confidence, whereas the ERN and its pMFC source are thought to also operate unconsciously (Charles et al. 2013). Consistent with this perspective, fMRI neural correlates of evidence against a choice were tracked in pMFC (the neural generator of the ERN), whereas more anterior prefrontal regions covaried with subjective confidence (Fleming et al. 2018).

An alternative perspective on the neural basis for broadcast and communication is provided by studies that have explicitly manipulated the requirement for a metacognitive judgment. For instance, one might compare trials on which a decision is made together with a metacognitive judgment of confidence against a control condition where the same kind of decision is made, but now the rating is about another property of the stimulus (e.g., its brightness or size). Such comparisons have highlighted a network of prefrontal regions, notably dorsal anterior cingulate cortex and lateral frontopolar cortex, in which activity is heightened when metacognitive judgments are required (Fleming et al. 2012, Qiu et al. 2018, Yeon et al. 2020).

A particularly detailed perspective on metacognitive judgment-related neural activation was provided by Gherman & Philastides (2018). Using EEG-informed fMRI, they could separate early neural activations correlating with confidence from later activations linked to the requirement for an explicit metacognitive judgment. Early confidence-related signals were seen in vmPFC [in a pgACC region similar to the one identified by Bang & Fleming (2018)], whereas later judgment-related activation was seen in lateral frontopolar cortex. Finally, in the study by Geurts et al. (2022) described above, the decoder's readout of sensory uncertainty in early visual areas was correlated with univariate fMRI signals in prefrontal cortex, consistent with domain-specific uncertainty estimates informing globally available estimates of propositional confidence.

An alternative approach to assaying the behavioral and neural signatures of broadcast and communication experimentally dissociates the private estimates of propositional confidence from the public estimates that are communicated to others. One natural way of achieving this is in a group context where individuals have to pool their confidence estimates to drive a group decision. Previous work has shown that when two individuals are collaborating in this way, the two partners rapidly and naturally adapt their confidence levels to converge on a common scale, so that one does not dominate the other (Bang et al. 2017). In an fMRI study of such social coordination about random dot motion judgments, it was found that whereas vmPFC (specifically, pgACC) covaried with private estimates of propositional confidence, as found in previous work, lateral frontopolar cortex additionally carried information about the extent to which a private-public mapping should be adjusted when communicating a public judgment (Bang et al. 2020). These findings are intriguing in light of other work emphasizing the role of frontopolar cortex in metacognitive efficiency (Allen et al. 2017, Baird et al. 2013, Fleming et al. 2010, McCurdy et al. 2013, Miyamoto et al. 2018). Such findings have often been interpreted as indicating a role for frontopolar cortex in supporting metarepresentations, with the impairment of the functions of this region leading to greater metacognitive noise. An alternative hypothesis is that frontopolar cortex constrains metacognitive efficiency by maintaining a stable private-public mapping, with instability in this mapping manifesting as a weaker coupling between metacognition and performance.

4.4. The Role of Self-Models

Up until now we have considered a relatively lean, minimal notion of propositional confidence, one that is directly informed by the internal states driving behavior (sometimes known as a

first-order model of confidence formation). However, a range of findings on human metacognition suggest that propositional confidence makes use of a richer (implicit) model of the factors affecting performance. The idea here is that, just as we build up a theory of how other minds work, we also build up a model of the factors affecting our own mental operations and bring that model to bear when making metacognitive judgments (Nelson & Narens 1990). Some of these background beliefs about how our minds work may be acquired via learning or be culturally inherited—as when children are instructed that feelings of fluency might produce misleading boosts in confidence, and they would be wise to slow down and reconsider their answer (Heyes et al. 2020). Differences between cultures in how these beliefs are acquired may account for findings of cultural differences in confidence and metacognition (van der Plas et al. 2022, Yates et al. 1998) and in how people process self-related feedback (Kitayama et al. 1997). Other background beliefs may be more innate and furnished by evolution, such as associations between interoceptive states and confidence (Allen et al. 2016, Fiacconi et al. 2016).

A long-standing proposal is that model-based contributions to metacognition rely on extensions of the models that guide our predictions of the mental states and behaviors of other people—a capacity known as theory of mind or mentalizing (Carruthers 2009). More generally, the implication is that we do not have direct access to first-order cognitive processes and instead have to infer their status from a variety of cues, just as we infer what others think or feel from observing their behavior. This view casts (model-based) metacognition as operating on similar principles to perception, in that both rely on the principles of (unconscious) inference.

A prominent theory in the metamemory literature proposes that a variety of cues affect metacognitive judgments via an inferential process. This renders metacognition susceptible to illusions and distortions, which are metacognitive analogues of perceptual illusions (Alter & Oppenheimer 2009). For instance, we may hold a belief that faster decisions are more likely to be accurate and use these feelings of fluency to inform our confidence estimates (Kiani et al. 2014). Similar boosts in fluency can be achieved by increasing the brightness of a face stimulus (Busey et al. 2000) or the font size of a word stimulus (Hu et al. 2015), leading to greater confidence in recall without any change in performance. Other work indicates that interoceptive factors influence confidence judgments even if they are irrelevant to the decision at hand (Fiacconi et al. 2016). For instance, Allen et al. (2016) found that subliminally presented disgusted faces not only led to changes in pupil dilation and heart rate but also modulated confidence in a perceptual (random dot motion) decision. The existence of these effects indicates the influence of an (implicit) self-model at work in the construction of explicit confidence judgments in a range of domains.

There has been relatively little work assaying the computational basis of model-based metacognitive inference or how such models are instantiated in the brain. One possibility is that self-models furnish beliefs about the parameters of the confidence formation process (which may not always match the actual parameters of such a process; Fleming & Daw 2017, Khalvati et al. 2021, Marcke et al. 2022). For instance, Hu et al. (2021) suggested that people's judgments of learning are constructed by integrating their processing experience on single trials with prior beliefs about how different cues affect memory performance—even if such cues do not promote objective success. Similarly, in the perceptual domain, Winter & Peters (2022) found that people misperceive sensory noise in the periphery of the visual field, leading to an inflation of perceptual confidence relative to perceptual acuity. This work implies a close connection between model-based influences on metacognition and the role of priors in propositional confidence formation. In an elegant experiment, Marcke et al. (2022) modulated people's priors on perceptual confidence through the use of false feedback on their relative scores compared to those of other participants. This influence was best captured by a model in which the parameters relating evidence accumulation to confidence were modified by a prior belief, without affecting objective accuracy or response times.

Effects of self-action on metacognitive judgments are another potential manifestation of model-based influences on confidence formation. Fleming & Daw (2017) proposed that a confidence computation may leverage information provided by one's own actions when inferring whether a decision is likely to be correct. Telltale signs of this effect have been confirmed empirically: Metacognitive sensitivity is often better when confidence judgments are provided after, compared to before, an explicit decision has been made (Pereira et al. 2020, Siedlecka et al. 2016, Wokke et al. 2020), with activity in frontopolar and insula cortex, and beta-band synchrony between motor and frontal cortex, hypothesized to mediate the impact of self-actions on metacognitive estimates (Pereira et al. 2020, Wokke et al. 2020). Conversely, metacognitive sensitivity is reduced when a task-relevant motor action is disrupted by applying transcranial magnetic stimulation over premotor cortex (Fleming et al. 2015). While these findings remain to be fully assimilated into computational models of confidence, they indicate that metacognitive judgments are sensitive to a range of internal cues that go beyond first-order performance.

More broadly, as noted above, one influential view is that model-based influences on metacognition may draw on similar resources to those supporting mentalizing about others. There is circumstantial evidence for this link, with similar developmental trajectories (both metacognition and mentalizing emerge around the age of 3–4) and overlap in neural correlates (particularly in the medial prefrontal cortex; Vaccaro & Fleming 2018). Recently, in an elegant series of studies, Nicholson and colleagues (2021) found that perceptual metacognitive sensitivity on a task requiring explicit confidence judgments (but not, intriguingly, one requiring an implicit gamble of the kind often used in animal metacognition experiments) correlated with mentalizing abilities and was impaired in subjects with autism spectrum disorder. In addition, a secondary mentalizing task (but not another, equivalently demanding task) interfered with explicit metacognitive judgments (Nicholson et al. 2021). Taken as a whole, this work suggests that the model-based component of human metacognition may co-opt social cognitive resources—although how such resources interface with the bottom-up aspects of propositional confidence formation reviewed above remains to be determined.

5. CONFIDENCE FORMATION AND THE PSYCHOLOGY OF METACOGNITION

We can now take stock of the discussion above and consider how these different computational stages interact and map onto the psychology of metacognition. First, myriad uncertainties exist at all stages of perception and cognition. Such uncertainties encompass not only well-studied perceptual systems but also internal uncertainties arising from memory, or uncertainties in interoception. Sensitivity to uncertainty is a central aspect of (first-order) Bayesian computation, but alone it is not evidence for metacognition. A further stage encodes confidence relative to a proposition—a (hypothetical) statement or decision—in a self-centered reference frame. This stage qualifies as (model-free) metacognition in that it has a mental state of the self (the proposition) among its correctness conditions (Carruthers & Williams 2022). A sensible agent will make use of domain-specific uncertainty when forming propositional confidence, and in some constrained scenarios, the latter will be a minor transformation of the former (consider, for instance, a posterior belief over potential motion directions that is transformed into propositional confidence in a specific choice option). It is therefore important to be aware that some tasks held to measure metacognition, such as the opt-out task, are often ambiguous with respect to whether they are tracking metacognitive (propositional) confidence or world-centered uncertainty. Propositional confidence can be globally broadcast and used in a range of metacognitive control functions, including strategically adjusting how confidence estimates are communicated to others. Finally, the formation of

propositional confidence may itself be influenced by an implicit model of how first-order cognitive systems operate.

These different stages can tentatively be mapped to systems-level interactions between brain areas. As noted above, early sensory areas may represent uncertainty over sensory variables such as motion direction, whereas prefrontal regions such as pmPFC and vmPFC track propositional confidence. Lateral frontopolar cortex is recruited to allow global broadcast and strategic communication of propositional confidence estimates. We can also advance the hypothesis that regions involved in theory of mind—including vmPFC, dorsomedial PFC, and temporoparietal junction—may support self-models that contribute to model-based metacognition (Vaccaro & Fleming 2018, Wittmann et al. 2016). However, it is important here to distinguish between propositional confidence in self-actions, propositional confidence in the actions of others, and the roles that models of self and other play in the formation of metacognitive judgments. Recent brain imaging studies suggest that propositional confidence formation in self and other draws on distinct brain networks (Bang et al. 2022, Jiang et al. 2022), but dorsomedial PFC may act as a common node for furnishing model-based information for both metacognition and mentalizing (Jiang et al. 2022, Wittmann et al. 2016).

Although up until now these components have been presented as distinct, this is for didactic convenience, and we should expect mutual interactions between them to be the norm. Indeed, the interrelationships between different stages of metacognitive computation are only just beginning to be investigated (Bang & Fleming 2018, Geurts et al. 2022, Shekhar & Rahnev 2018), but understanding them represents a major goal for the field (Rahnev et al. 2022). One possibility is that neural codes within different frames of reference emerge and are maintained in parallel, serving different computational goals. For instance, evidence may be accumulated about particular sensory features (world-centered frame of reference) and, simultaneously, about (future or past) choice correctness (self-centered frame of reference), with the latter feeding back to set the bound on current or future sensory evidence accumulation (Balsdon et al. 2020). As such, it is likely to be more fruitful to view metacognition as emerging from a set of dynamically interacting internal states, some of which are world-centered while others encode beliefs about one's propositions or decisions (Yeung & Summerfield 2012).

6. REVISITING CURRENT CONTROVERSIES

6.1. Biases and Suboptimalities in Confidence

A long-running debate in the field is between those who consider confidence (and, by implication, metacognition) to be inherent to the decision process, and those who consider it to depend on additional machinery or computation. In emphasizing multiple computational components, the current framework provides a resolution of this tension. In certain scenarios, such as the decision to opt out of a well-constrained decision problem, propositional confidence can be derived from a direct transformation of the accumulated evidence for one choice or the other (Kiani & Shadlen 2009, van den Berg et al. 2016a). However, in other scenarios—for instance, when postdecisional accumulation of evidence is at play or when there are multiple model-based cues to confidence—a second stage of propositional confidence formation may be involved, particularly when it is functionally advantageous to broadcast such confidence to multiple distinct consumer systems. In that situation, dedicated machinery for the readout and usage of propositional confidence (for instance, in PFC) may be the norm rather than the exception, with lesions or damage to these downstream areas manifesting as selective metacognitive deficits and presenting more opportunity for deviations from ideal observer models to occur.

A fruitful approach to pursuing the computational basis of metacognition, then, is to explicitly model these different stages and ask how noise or suboptimalities within each component may contribute to metacognitive inefficiencies (Guggenmos 2022, Mamassian & de Gardelle 2022, Shekhar & Rahnev 2021). For instance, Boundy-Singer et al. (2023) identify meta-uncertainty about sensory uncertainty as a key domain-general constraint on the fidelity of propositional confidence estimates in both perceptual and cognitive decision tasks. In turn, constraints on post-decisional evidence accumulation may affect the extent to which confidence estimates faithfully track performance (Desender et al. 2022, Pleskac & Busemeyer 2010). When interacting with others, there is a requirement to maintain distinct models for ourselves and others, and selecting the correct model may be computationally demanding: Wittmann et al. (2016) found that when tracking the performance of oneself and others, people sometimes merged their feedback with that of others. This intertwining of models of self- and other-performance was associated with differences in activity in dorsomedial prefrontal cortex, and disrupting this area using transcranial magnetic stimulation (TMS) led to greater self-other mergence (Wittmann et al. 2021)—suggesting that one function of this brain region is not only to support models of ourselves and others but also to keep these models apart. More generally, different suboptimalities may coexist, and the same kind of computational constraints that affect first-order cognition are likely to affect the suboptimality of metacognition (Rahnev & Denison 2018).

One metacognitive bias that has received particularly detailed theoretical and empirical scrutiny is the positive evidence bias (PEB). The PEB manifests as confidence being more affected by evidence in favor of a choice than by evidence against it (Zylberberg et al. 2012), such that an increase in overall evidence results in boosts in confidence even though performance remains unaffected. Initial theoretical explanations proposed that the PEB may result from a bias in the broadcast or readout of propositional confidence estimates, or from a heuristic applied to evidence spaces that are often detection- rather than discrimination-like (Maniscalco et al. 2021, Miyoshi & Lau 2020). More recently, though, empirical and modeling studies have led to surprising conclusions that constrain the origins of the PEB. First, a PEB emerges within a convolutional neural network that is trained to both discriminate digits and estimate confidence in these classifications—indicating that a PEB may be not a foible of human metacognition but rather a core feature of how high-dimensional evidence spaces are mapped to propositional confidence (Webb et al. 2023). Second, the PEB can be flipped, creating a negative evidence bias, if the decision is reframed as a search for the weaker response option (e.g., fewer dots or a disliked item; Sepulveda et al. 2020). Together, these findings point toward a model in which the PEB may be a feature of how propositional confidence is formed, rather than a bias in the tracking of domain-specific uncertainties (Mazor et al. 2023).

6.2. Sources of Domain-Generality

Another contested issue is the extent to which metacognitive capacities should be considered domain-general or domain-specific. Behaviorally, individual differences in metacognitive efficiency have been shown to be correlated across distinct task domains, after controlling for correlations in performance (Ais et al. 2016, Faivre et al. 2018, Mazancieux et al. 2020, Rouault et al. 2018b). However, the strength of these correlations is often weak and variable, especially in the smaller samples used in neuroimaging research (for a meta-analysis, see Rouault et al. 2018b). In addition, there are concerns that factors only indirectly related to metacognitive capacity may contribute to findings of domain-generality—such as how confidence scales are used, or the adoption of a particular threshold for postdecisional evidence accumulation (Desender et al. 2022, Xue et al. 2021). Findings of domain-generality in metacognitive bias (average confidence level) are

INDIVIDUAL AND GROUP DIFFERENCES

Metacognitive efficiency shows moderate test-retest reliability, both across different sessions of the same experiment (Ais et al. 2016, Fleming et al. 2010) and across different days (Wright et al. 2012). Metacognitive bias (calibration) shows stronger test-retest reliability, with stable confidence fingerprints seen across different tasks and testing sessions (Ais et al. 2016). A number of studies have linked local and global metacognitive biases to individual differences in transdiagnostic mental health symptoms, including anxiety, depression, self-esteem, and compulsivity (Hoven et al. 2019, Seow et al. 2021). Conversely, metacognitive efficiency is predictive of individual differences in dogmatism about real-world issues such as politics and climate change (Fischer et al. 2019, Rollwage et al. 2018), with the parameters governing confidence-based control correlating with people's openness to new information (Schulz et al. 2020).

more robust and have been related to features of both personality and mental health (see the sidebar titled Individual and Group Differences).

Set against findings of correlated individual differences are findings of both domain-specificity in the neural basis of metacognition and domain-specific impairments in metacognitive efficiency following lesions or experimental intervention. One particularly consistent set of findings points to a selective role for medial parietal cortex (precuneus) in metamemory (Baird et al. 2013, McCurdy et al. 2013). Accordingly, lesions to frontopolar cortex (but not precuneus) impair metaperceptual efficiency but leave metamnemonic efficiency (as assayed by recognition memory confidence) intact (Fleming et al. 2014). The reverse dissociation is seen with theta burst TMS to the precuneus, which impairs metamnemonic but not metaperceptual efficiency (Ye et al. 2018).

The current framework provides an opportunity to integrate these findings. A positive manifold of individual differences in healthy metacognition may be mediated by common downstream processes involved in the formation of propositional confidence and/or its global broadcast. Conversely, domain-specific limitations may be imposed by how domain-specific uncertainty is propagated into a propositional confidence computation and/or the fidelity of model-based estimates of uncertainty parameters (i.e., uncertainty about uncertainty; Boundy-Singer et al. 2023, Khalvati et al. 2021). For instance, one plausible, although speculative, role for precuneus in metamemory is that it is involved in translating uncertainty-related information carried by hippocampal neurons (Rutishauser et al. 2015) into a (prospective or retrospective) propositional confidence judgment. Understanding these interactions will be aided by new data analysis approaches that seek to understand which variables can be easily read out from mixed selectivity neural populations—with the possibility that both domain-general and domain-specific components of confidence formation coexist within the same brain area (Fu et al. 2022, Morales et al. 2018). At a behavioral level, future work should seek to move beyond examining correlations in descriptive statistics such as meta- d' and instead seek to characterize the computational stages at which domain-general in metacognition emerges (Boundy-Singer et al. 2023, West et al. 2023).

7. WHERE NEXT?

7.1. Searching for Common Computational Principles

As indicated in the preceding section, a key next step is to move beyond the useful but artificial division of metacognition research into distinct domains, such as memory or perception, in order to characterize common computational principles that constrain metacognitive capacity. In this respect, the often segregated fields of perceptual and memory metacognition research can

learn a lot from each other, and further cross-fertilization will no doubt reap benefits. For instance, the metaperception field has developed psychophysical paradigms that permit the detailed computational modeling of pre- and postdecisional processes, in which hundreds of trials per participant are often required to fully characterize the joint distribution of accuracy, response time, and confidence. These endeavors have been accelerated by the development of the Confidence Database and the adoption of consensus goals in the field (Rahnev et al. 2020, 2022). Conversely, the metamemory field has tended to leverage more naturalistic stimuli (e.g., memory for faces) and developed clever experimental designs to carefully unpack the contribution that a range of cues make to metacognitive judgments (e.g., the illusory boosts in confidence that ensue from manipulations of fluency).

7.2. From Local to Global Metacognition

Most research on propositional confidence has focused on local judgments of performance on individual trials or task episodes. In contrast, a distinct literature in social and clinical psychology has focused on how people evaluate themselves at a global level—for instance, their self-efficacy, or estimates of their abilities relative to others. These global self-evaluations are related to future attainment (via an impact on motivation and task engagement) and may govern adaptive behavior such as knowing when to seek help or offload to the environment. However, little is known about how local metacognitive computations influence and shape self-evaluations over this longer timescale.

One fruitful approach considers global confidence as a higher-order prior on estimates of (local) propositional confidence (Boldt et al. 2019, Marcke et al. 2022), which can be naturally modeled as a probability distribution over expected success (Rouault et al. 2019). In the absence of any local task experience, people access this prior when making confidence judgments—for instance, estimating the chances they will score from a free kick (see **Figure 1**). In turn, this prior can be updated in light of local (retrospective) confidence in individual actions or decisions. Tentative evidence for this view comes from experiments in which subjects provided intermittent global confidence estimates on a perceptual task (Lee et al. 2021, Rouault et al. 2019). Global confidence was informed by local confidence fluctuations during the previous block, and using fMRI, it was found that vmPFC and precuneus integrate local confidence over longer timescales to track aggregate self-performance (Rouault & Fleming 2020, Wittmann et al. 2016).

Another perspective on how propositional confidence unfolds over longer timescales is provided by studies that have examined how subjects estimate the probability of making task errors based on recent experience. In an elegant paradigm, Purcell & Kiani (2016) found that subjects track a prior on propositional confidence by integrating evidence over multiple trials, and they leverage this prior on expected task accuracy to decide whether to switch strategy (in effect, reaching a threshold at which they decide to blame the error on the task rather than on themselves). Neurons in monkey pmPFC were found to integrate information about previous trials and drive decisions about whether to switch strategy (Sarafyazd & Jazayeri 2019). Similarly, in human pmPFC, neuronal populations signal expected conflict probability (a proxy for propositional confidence) across trials as a state variable that is orthogonal to within-trial dynamics, just as might be expected for neural activity encoding a prior on confidence level (Fu et al. 2022).

7.3. Opportunities for Metacognitive Interventions

Interventions to modify metacognition are in their infancy, but here, too, progress could benefit from understanding which computational stages are being affected. Previous work has suggested that metacognitive efficiency may be modulated in response to meditation (Baird et al. 2014), drugs

(Hauser et al. 2017), neurofeedback (Cortese et al. 2016), brain stimulation (Shekhar & Rahnev 2018), and training (Carpenter et al. 2019). However, with some notable exceptions (Shekhar & Rahnev 2018), the locus of action of these effects remains poorly understood. Knowing which steps in a computational chain are targeted by an intervention helps to identify how and whether metacognitive boosts are likely to generalize beyond the lab as well as what functional benefits they might provide. For instance, a beta blocker may inhibit the contribution of model-based interoceptive cues to confidence estimates, therefore improving metacognitive efficiency on a constrained laboratory task but impairing it in situations in which those cues are more valid. As another example, delivering feedback to improve confidence calibration over a period of two weeks shows promise in elevating metacognitive efficiency not only on the trained task but also more broadly (Carpenter et al. 2019). However, recent work suggests that the incentives underpinning this intervention primarily acted upon the way that confidence was communicated via a confidence scale (Rouy et al. 2022)—at the level of a private-public mapping, rather than at the level of propositional confidence formation. Such an intervention may still be useful in social situations where public confidence estimates are being pooled across observers but would be less useful in cases in which propositional confidence is being used for intrapersonal control.

8. CONCLUSIONS

The fields of metacognition and confidence research are natural allies but have often been uneasy bedfellows. Here I argue that metacognition research is the study of propositional confidence in all its forms. Once this is recognized, it opens up the problem of how different computational components of confidence formation interact, including those supporting the rich self-models that humans bring to bear when evaluating their behaviors and internal states. In this endeavor, the different subfields of metacognition research have a lot to learn from each other. There is no reason to think that representations of uncertainty are any less relevant for understanding metamemory, or that the contribution of self-models and other heuristics is less relevant for understanding perceptual confidence. A research program that bridges this divide, and which seeks to understand the full range of computational stages underpinning human metacognition, will likely benefit from the lessons that can be gleaned from both of these literatures. In turn, disparate findings on the neural basis of uncertainty and performance monitoring can be integrated into a common framework, and a new understanding of the locus of action of metacognitive interventions can be achieved.

SUMMARY POINTS

1. Confidence research has focused on subpersonal representations of confidence and uncertainty in sensory or motor tasks.
2. Metacognition research is concerned with personal-level beliefs about performance.
3. These viewpoints can be reconciled by recognizing metacognitive judgments as propositional confidence estimates about (hypothetical) decisions or actions.
4. A key step in forming propositional confidence is shifting between world- and self-centered frames of reference when encoding uncertainty.
5. Propositional confidence can be globally broadcast to support a range of metacognitive control functions, including social communication.
6. Model-based influences on confidence formation (such as beliefs and priors about performance) may share parallels with theory of mind.

FUTURE ISSUES

1. What are the common computational principles that constrain metacognitive capacity across domains?
2. How can computational models of confidence formation capture model-based influences on metacognition?
3. Do model-based influences on metacognition share neural and computational resources with theory of mind?
4. Can models of confidence developed in psychophysical experiments be generalized to naturalistic scenarios?
5. Can novel metacognitive interventions be developed based on a refined understanding of the computational components of confidence?

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